

STQM 270 Final Project

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STQM 270: Introduction to Data Mining

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Unsupervised Clustering

Objectives

For the Unsupervised Clustering model, we used the Iris.xlsx dataset to configure multiple clustering models in IBM SPSS Modeler. The goal of this section was to group the Iris flower records into clusters based on their measurement variables without using the species name as a target variable. The model used four main input fields: sepal length, sepal width, petal length, and petal width. By comparing the results of K-Means, TwoStep, TwoStep-AS, and Kohonen clustering, we were able to evaluate how well each model separated the data into meaningful groups.

Description

The Iris dataset contains measurements for three types of Iris flowers: Setosa, Versicolor, and Virginica. Since this section uses unsupervised learning, the model does not try to predict a specific target. Instead, it looks for natural patterns in the data. The species number was filtered out so that the clustering models would rely only on the flower measurements. The data was also standardized using MinMax values so the variables could be compared on the same scale. This is important because clustering models are distance-based, and variables with larger ranges can otherwise have too much influence on the results.

Data Mining Session

Preview from Iris.xlsx Node (5 fields, 10 records)

File Edit Generate

Table Annotations

	Petal_width	Petal_length	Sepal_width	Sepal_length	Species_name
1	2.000	14.000	33.000	50.000	Setosa
2	2.000	10.000	36.000	46.000	Setosa
3	2.000	16.000	31.000	48.000	Setosa
4	1.000	14.000	36.000	49.000	Setosa
5	2.000	13.000	32.000	44.000	Setosa
6	2.000	16.000	38.000	51.000	Setosa
7	2.000	16.000	30.000	50.000	Setosa
8	4.000	19.000	38.000	51.000	Setosa
9	2.000	14.000	30.000	49.000	Setosa
10	2.000	14.000	36.000	50.000	Setosa

Preview of Iris dataset

Set Globals

Settings Annotations

Globals to be created:

Field	MEAN	SUM	MIN	MAX	SDEV
Petal_width	☐	☐	☒	☒	☐
Petal_length	☐	☐	☒	☒	☐
Sepal_width	☐	☐	☒	☒	☐
Sepal_length	☐	☐	☒	☒	☐

Default operation(s): ☐ MEAN ☐ SUM ☒ MIN ☒ MAX ☐ SDEV

A Global node was configured to calculate minimum and maximum values for each numerical field, including petal width, petal length, sepal width, and sepal length. These values were later used for standardization.

Field	Filter	Field
Species_No		Species_No
Petal_width		Petal_width
Petal_length		Petal_length
Sepal_width		Sepal_width
Sepal_length		Sepal_length
Species_name		Species_name

The species number field was filtered out to prevent it from influencing the clustering process.

Field	Measurement	Values	Missing	Check	Role
Petal_width	Continuous	[1.0,25.0]		None	Input
Petal_length	Continuous	[10.0,69.0]		None	Input
Sepal_width	Continuous	[20.0,44.0]		None	Input
Sepal_length	Continuous	[43.0,79.0]		None	Input
Species_name	Nominal	"Setosa","V...		None	Input

The species number field was filtered out to prevent it from influencing the clustering process. Each measurement variable was confirmed as continuous, while species name remained a nominal field. This step ensured the dataset was properly structured for unsupervised learning.

MINMAX

➡

➡

➡

Derive as: Formula

?

☐

☐

Settings

Annotations

Mode: ☐ Single ☒ Multiple

Derive from:

⌵

Petal_width

⌵

Petal_length

⌵

Sepal_width

⌵

⌵

⌵

⌵

Field name extension: Add as: ☒ Suffix ☐ Prefix

Derive as:

Formula

TIP: Refer to selected fields by using @FIELD

Field type:

⚡ <Default>

Formula:

1

(@FIELD-@GLOBAL_MIN(@FIELD)) / (@GLOBAL_MAX(@FIELD)-@GLOBAL_MIN(@FIELD))

⌵

⌵

A MinMax Derive node was used to standardize the dataset. Each measurement variable was transformed using a formula that rescales values between 0 and 1 based on the global minimum and maximum. This standardization ensures that all variables contribute equally to the clustering process, preventing variables with larger scales from dominating the model.

Type

Preview

Types

Format

Annotations

Read Values

Clear Values

Clear All Values

Field	Measurement	Values	Missing	Check	Role
Petal_width	Continuous	[1.0,25.0]		None	Input
Petal_length	Continuous	[10.0,69.0]		None	Input
Sepal_width	Continuous	[20.0,44.0]		None	Input
Sepal_length	Continuous	[43.0,79.0]		None	Input
Species_name	Nominal	"Setosa", "..."		None	Input
Petal_width_MINMAX	Continuous	[0.0,1.0]		None	Input
Petal_length_MINMAX	Continuous	[0.0,1.0]		None	Input
Sepal_width_MINMAX	Continuous	[0.0,1.0]		None	Input
Sepal_length_MINMAX	Continuous	[0.0,1.0]		None	Input

A Type node was applied to verify and assign the correct measurement levels to each field. All standardized variables were set as continuous inputs.

Filter

Preview

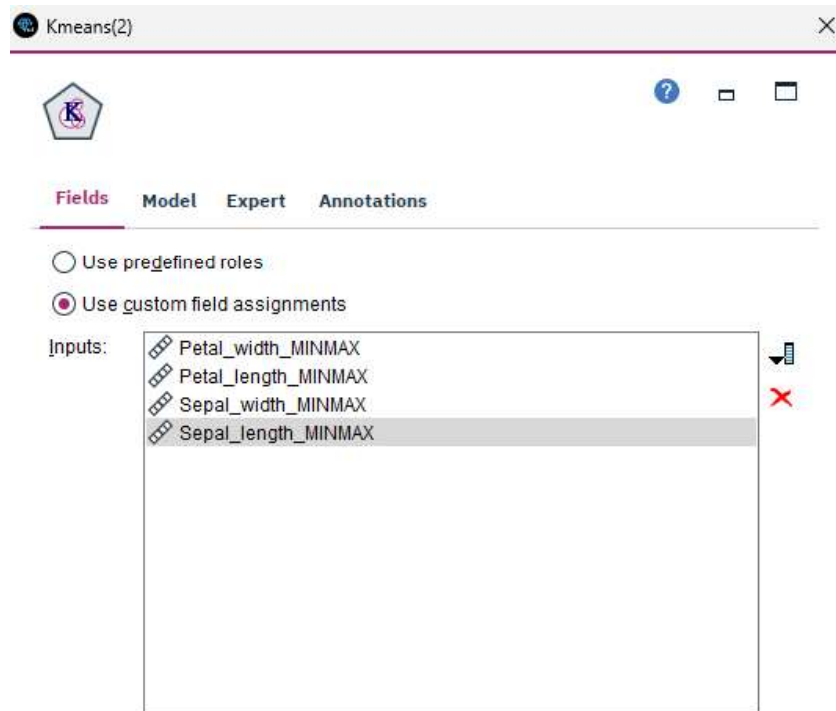
Filter

Annotations

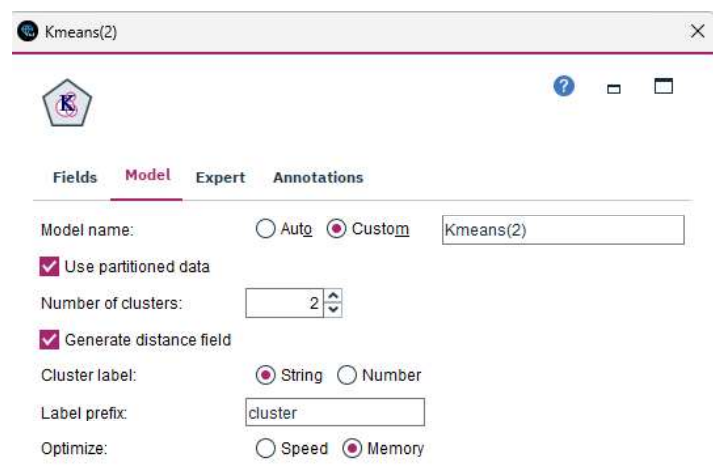
Fields: 9 in, 4 filtered, 0 renamed, 5 out

Field	Filter	Field
Petal_width		Petal_width
Petal_length		Petal_length
Sepal_width		Sepal_width
Sepal_length		Sepal_length
Species_name		Species_name
Petal_width_MINMAX		Petal_width_MINMAX
Petal_length_MINMAX		Petal_length_MINMAX
Sepal_width_MINMAX		Sepal_width_MINMAX
Sepal_length_MINMAX		Sepal_length_MINMAX

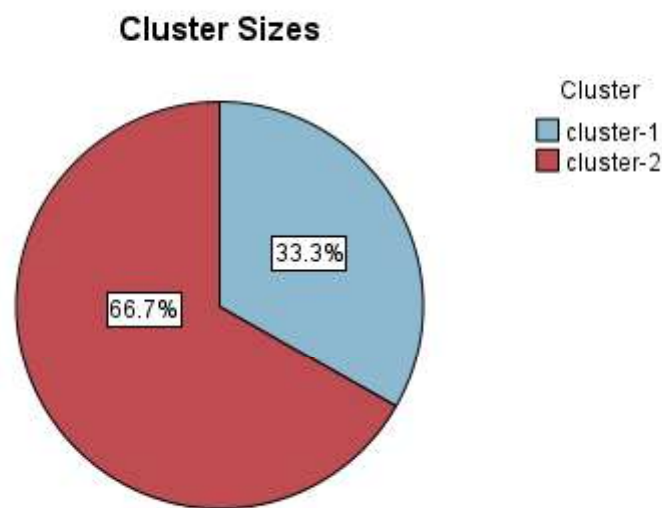
A Filter node was then used to remove the original unstandardized variables and retain only the MinMax standardized fields. This ensured that the clustering models would operate solely on normalized data.



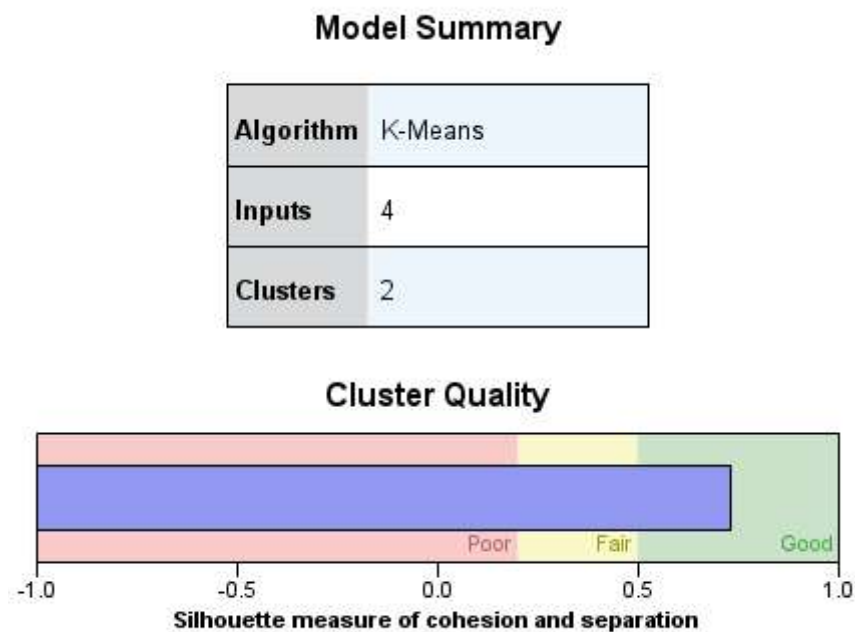
The K-Means model was configured using custom field assignments. The standardized variables petal width, petal length, sepal width, and sepal length were selected as input fields. No target variable was used, as this is an unsupervised model. The configuration prepared the dataset for clustering into two groups.



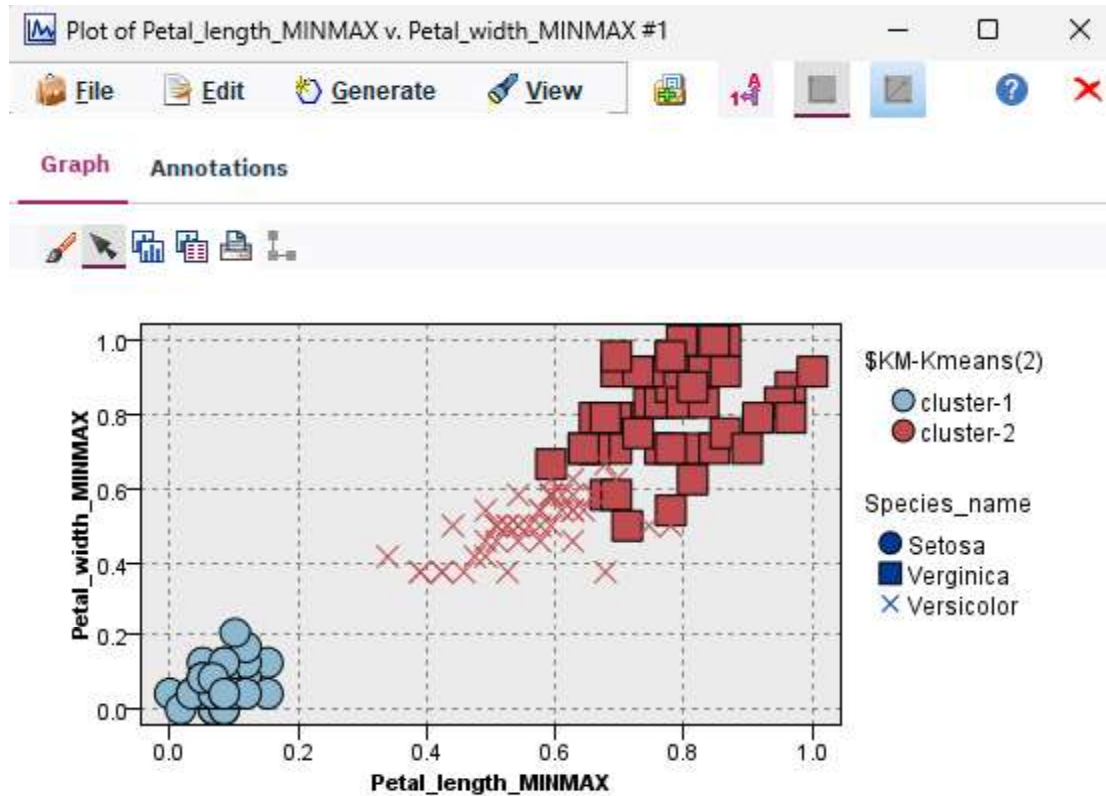
The K-Means model was set to generate two clusters.



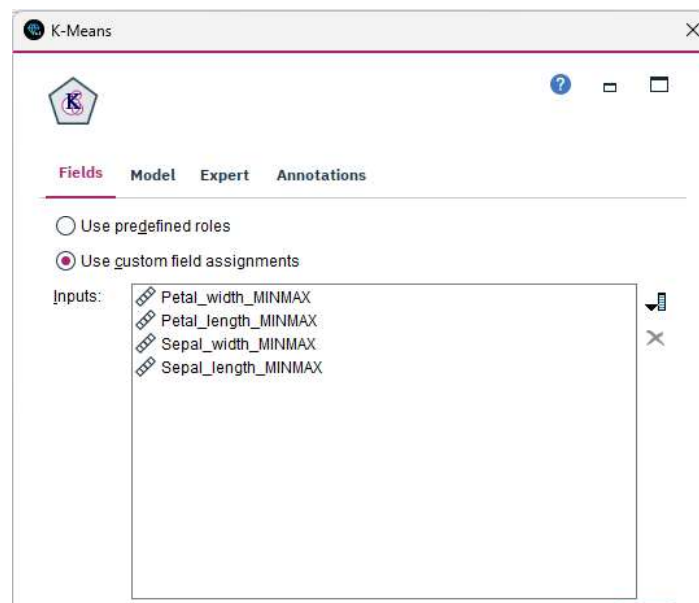
After running the model, a cluster size chart showed the distribution of records between the two clusters. Approximately two-thirds of the data fell into one cluster, while one-third fell into the other. This indicates an uneven but meaningful grouping within the dataset.



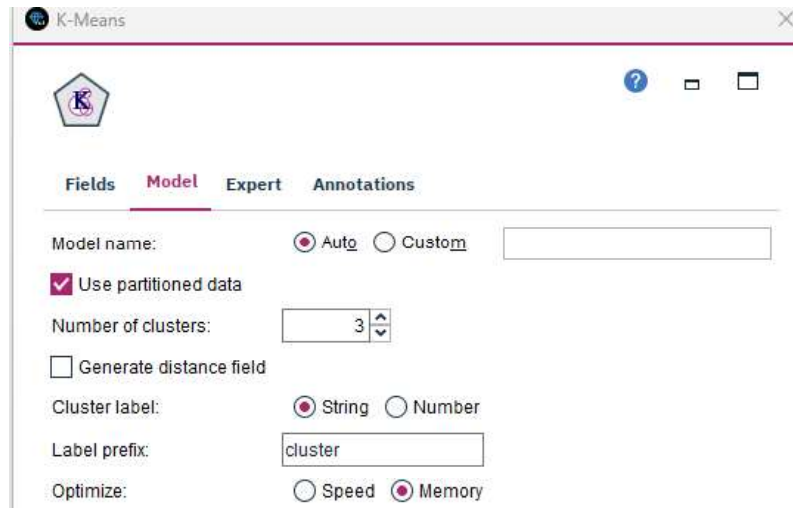
The model summary confirmed that the K-Means algorithm used four input variables and produced two clusters. The cluster quality measure indicated fair separation between clusters.



A scatterplot of petal length versus petal width showed that one cluster was clearly separated, likely representing the Setosa species, while the other cluster contained overlapping points from the remaining species.



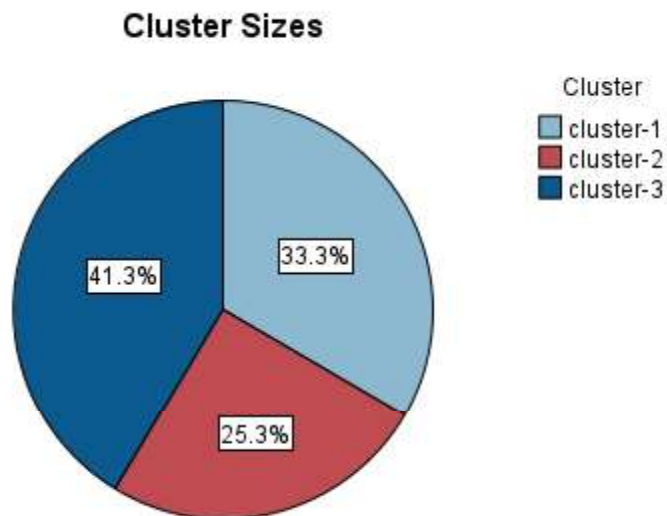
A second K-Means model was created with three clusters to better reflect the three species in the Iris dataset.



The screenshot shows the 'K-Means' model configuration window. The 'Model' tab is selected. The configuration includes:

- Model name: ☐ Auto ☐ Custom
- ☒ Use partitioned data
- Number of clusters:
- ☐ Generate distance field
- Cluster label: ☒ String ☐ Number
- Label prefix:
- Optimize: ☐ Speed ☒ Memory

The same standardized variables were used as inputs. The model configuration was adjusted to specify three clusters instead of two

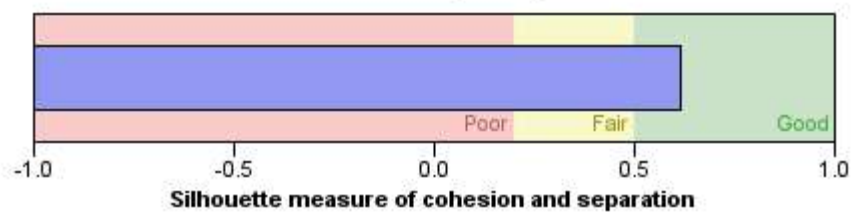


The K-Means model with three clusters produced a more balanced distribution of data across clusters. The cluster size chart showed three distinct groups.

Model Summary

Algorithm	K-Means
Inputs	4
Clusters	3

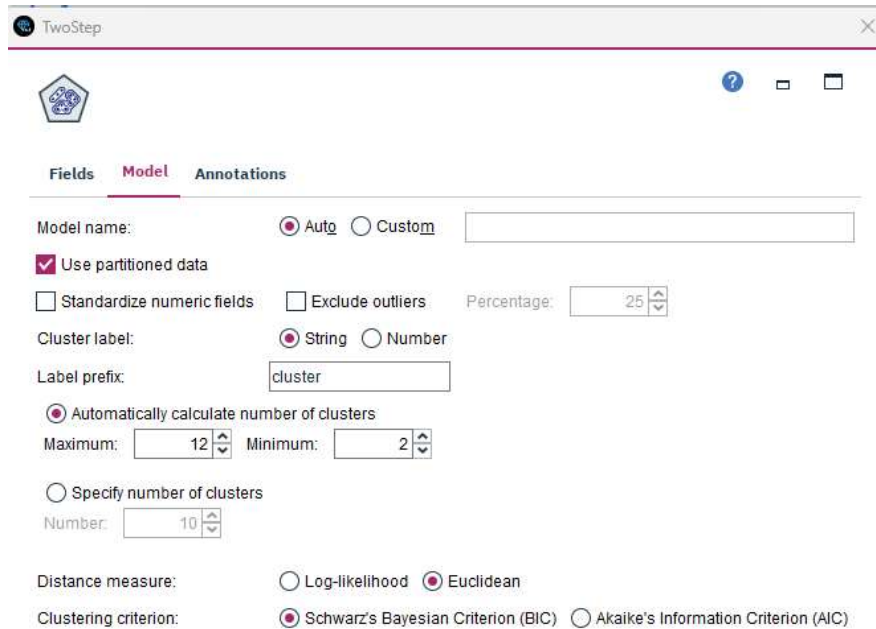
Cluster Quality



The model summary confirmed the use of four inputs and three clusters. The cluster quality remained fair, suggesting moderate separation between the groups



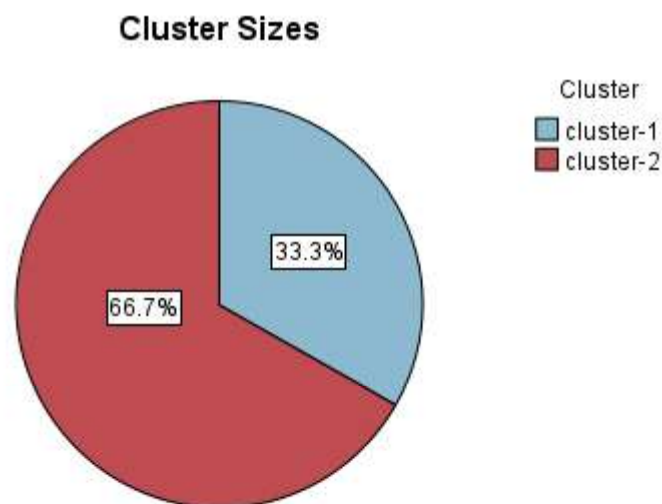
The TwoStep clustering model was configured using the same standardized input variables.



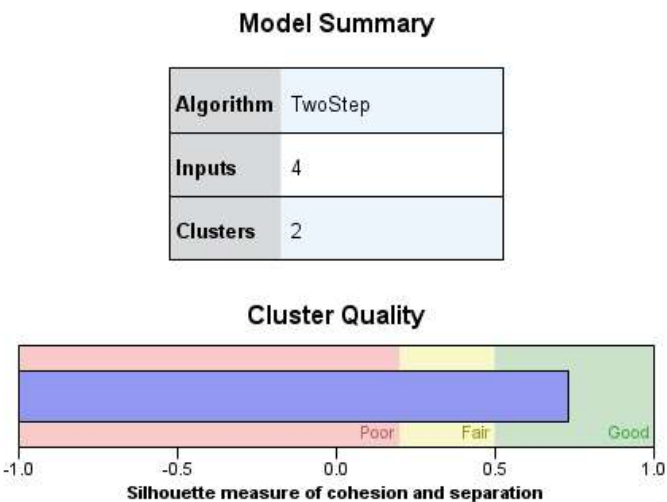
The screenshot shows the 'TwoStep' software window with the 'Model' tab selected. The interface includes a logo, a help icon, and window controls. The 'Model' tab has three sub-tabs: 'Fields', 'Model', and 'Annotations'. The 'Model' sub-tab contains the following settings:

- Model name: ☐ Auto ☐ Custom
- ☒ Use partitioned data
- ☐ Standardize numeric fields ☐ Exclude outliers Percentage:
- Cluster label: ☒ String ☐ Number
- Label prefix:
- ☒ Automatically calculate number of clusters
Maximum: Minimum:
- ☐ Specify number of clusters
Number:
- Distance measure: ☐ Log-likelihood ☒ Euclidean
- Clustering criterion: ☒ Schwarz's Bayesian Criterion (BIC) ☐ Akaike's Information Criterion (AIC)

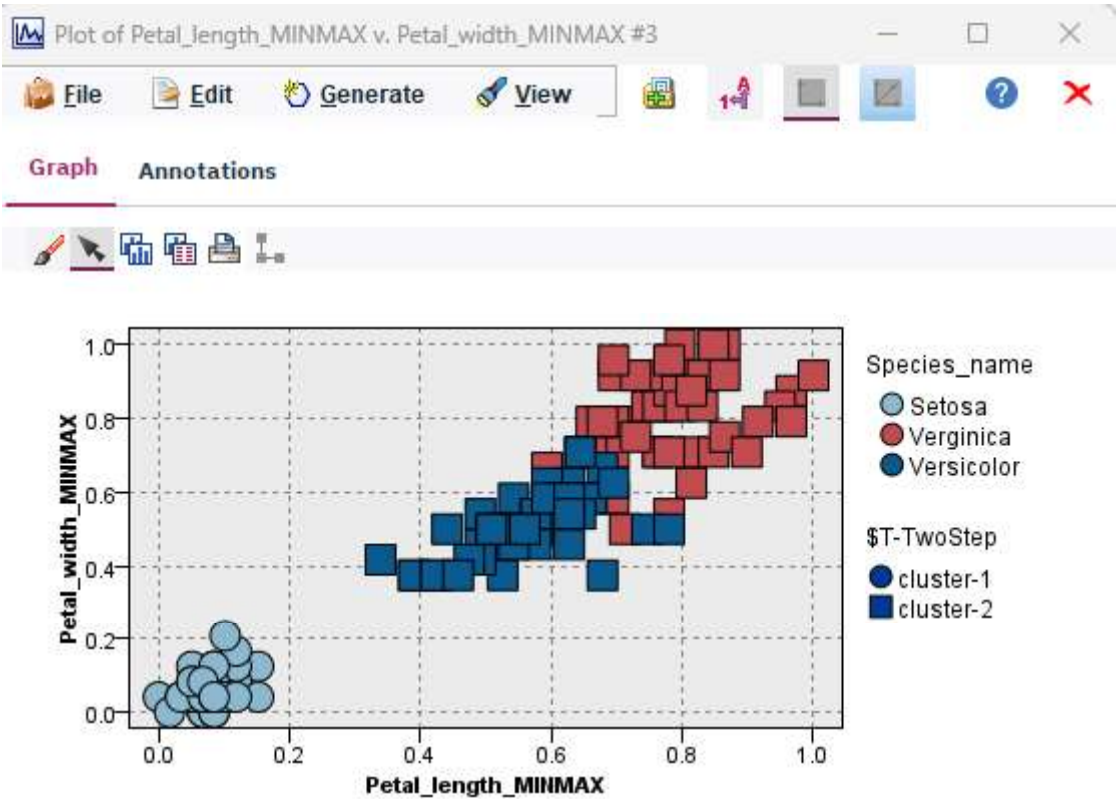
Custom field assignments were used, and the model was set to automatically determine the optimal number of clusters. This method combines hierarchical and partitioning approaches for clustering.



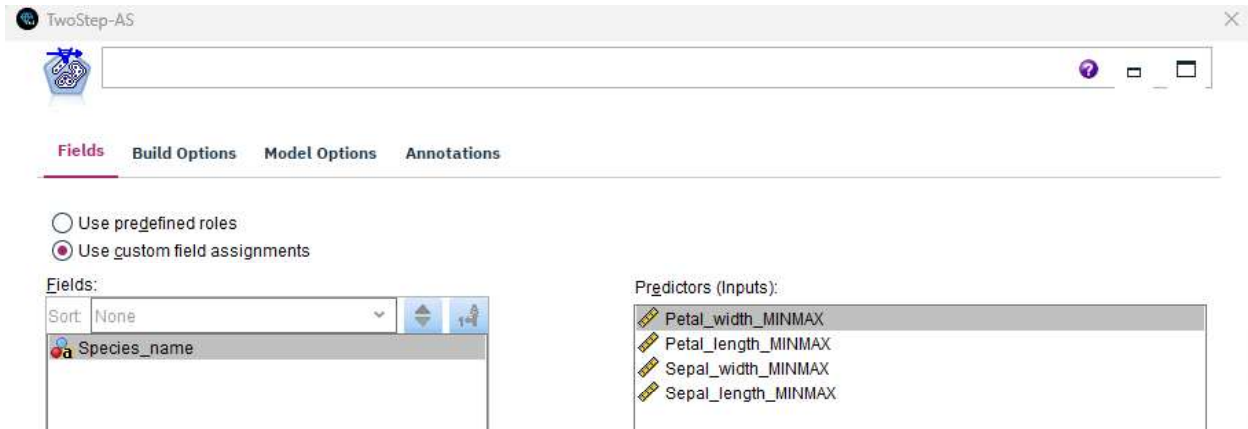
The TwoStep model produced two clusters. The cluster size chart showed that one cluster contained approximately one-third of the data, while the other contained about two-thirds.



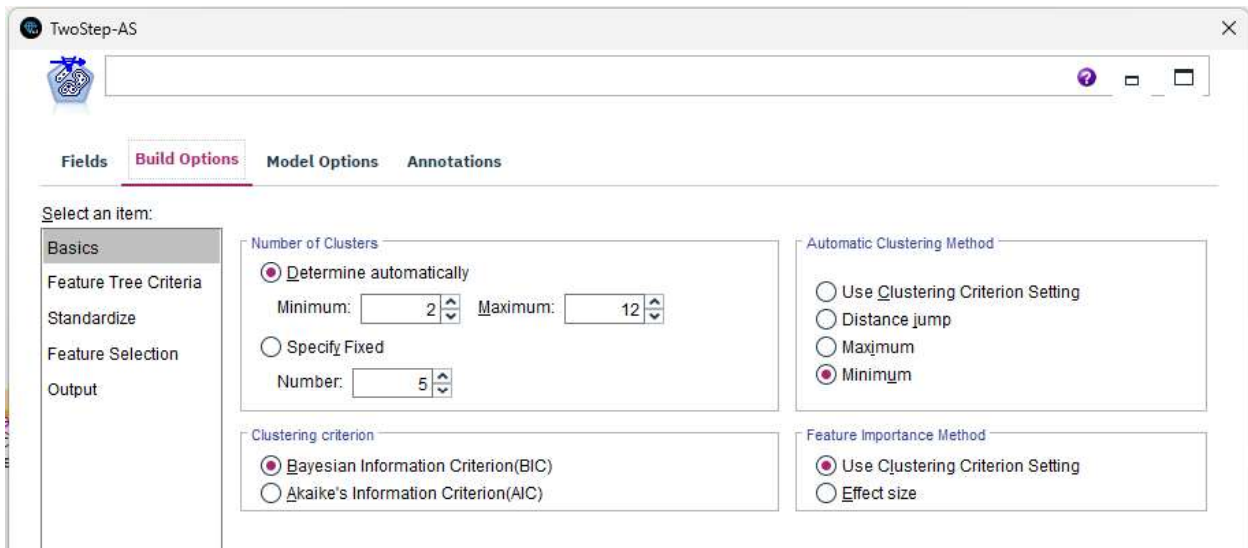
The model summary confirmed the algorithm and input variables. The cluster quality measure indicated fair separation.



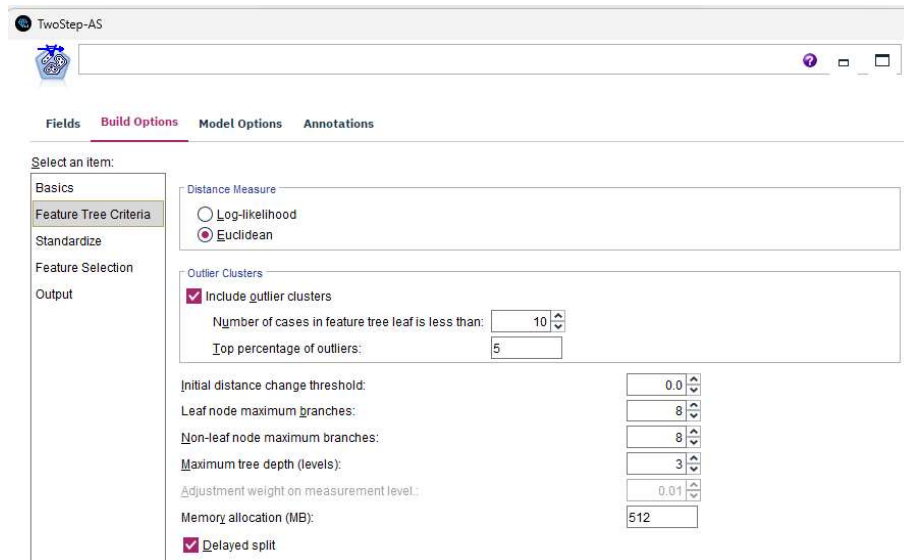
A scatterplot of standardized petal length and petal width showed the cluster distribution visually.



The TwoStep-AS model was then configured by assigning the species name as an additional field for comparison purposes. This allowed for the evaluation of how well the clusters aligned with actual species.

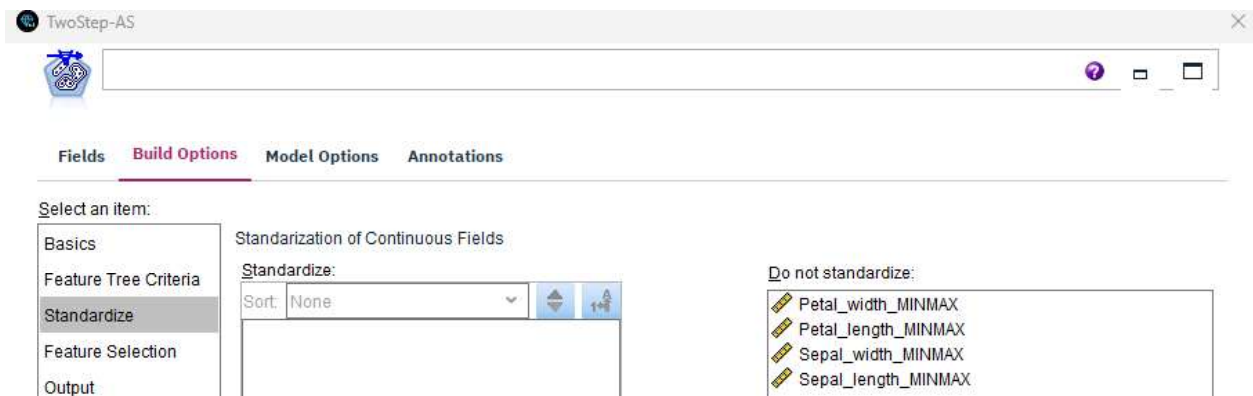


The TwoStep-AS model was configured with a minimum of two clusters and a maximum of twelve clusters. The number of clusters was set to five for evaluation.



The screenshot shows the 'TwoStep-AS' application window with the 'Build Options' tab selected. On the left, a sidebar lists 'Select an item:' with categories: Basics, Feature Tree Criteria, Standardize, Feature Selection, and Output. The 'Distance Measure' section has two radio buttons: 'Log-likelihood' (unselected) and 'Euclidean' (selected). The 'Outlier Clusters' section has a checked checkbox 'Include outlier clusters'. Below it, 'Number of cases in feature tree leaf is less than:' is set to 10, and 'Top percentage of outliers:' is set to 5. Further down, several numerical settings are shown in input boxes: 'Initial distance change threshold' (0.0), 'Leaf node maximum branches' (8), 'Non-leaf node maximum branches' (8), 'Maximum tree depth (levels)' (3), 'Adjustment weight on measurement level' (0.01), and 'Memory allocation (MB)' (512). At the bottom, the 'Delayed split' checkbox is checked.

Euclidean distance was selected as the distance measure. These settings allowed the model to determine optimal groupings based on statistical measures.



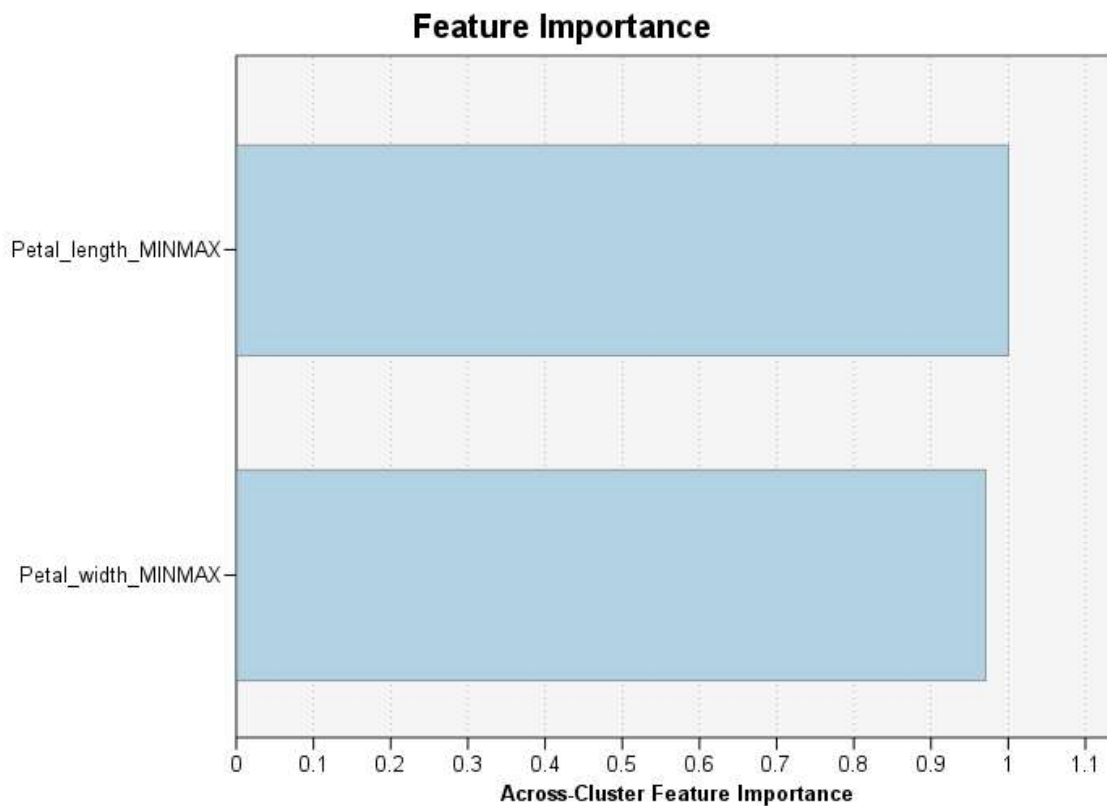
The screenshot shows the 'TwoStep-AS' application window with the 'Build Options' tab selected. The 'Standardize' menu item in the sidebar is highlighted. The 'Standardization of Continuous Fields' section is active. Under 'Standardize:', there is a 'Sort:' dropdown menu set to 'None' and a list of fields to be standardized. To the right, under 'Do not standardize:', a list of fields is shown, each preceded by a yellow eraser icon, indicating they are excluded from standardization: 'Petal_width_MINMAX', 'Petal_length_MINMAX', 'Sepal_width_MINMAX', and 'Sepal_length_MINMAX'.

Standardize menu for twostep AS and make sure that all the values are in the do not standardize section

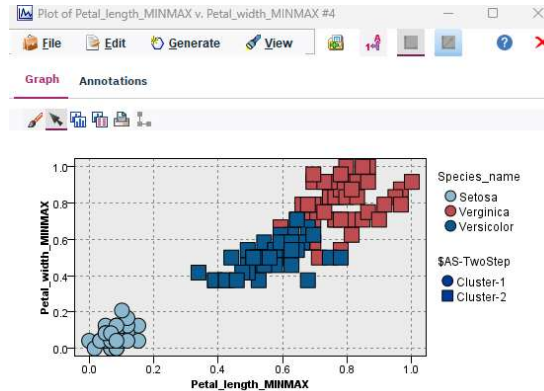
TwoStep-AS**Model Information**

Model Specifications			
Minimum Number of Regular Clusters		2	
Maximum Number of Regular Clusters		12	
Adaptive Feature Selection		On	
Feature Importance Method		Information Criterion	
Information Criterion		Bayesian Information Criterion (BIC)	
Distance Measure		Euclidean	
Final Model	Number of Regular Clusters	2	
	Number of Outlier Clusters	0	
	Continuous Inputs	Petal_width_MINMAX Petal_length_MINMAX	

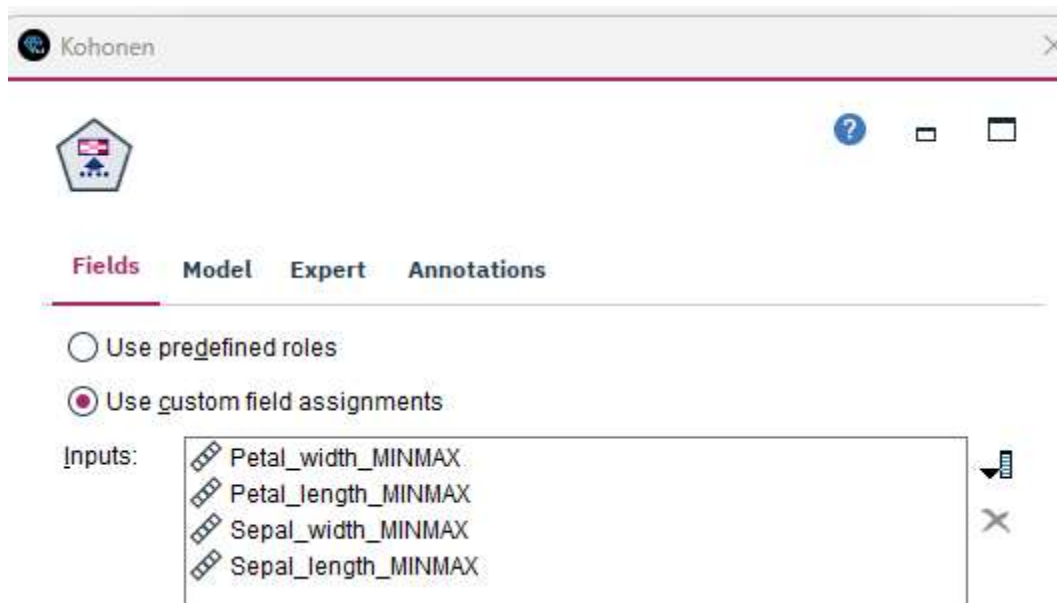
The model information showed that the TwoStep-AS model produced two clusters. It confirmed that Euclidean distance was used and that the clustering criterion was based on the Bayesian Information Criterion. The model also showed that no outliers were detected in the dataset.



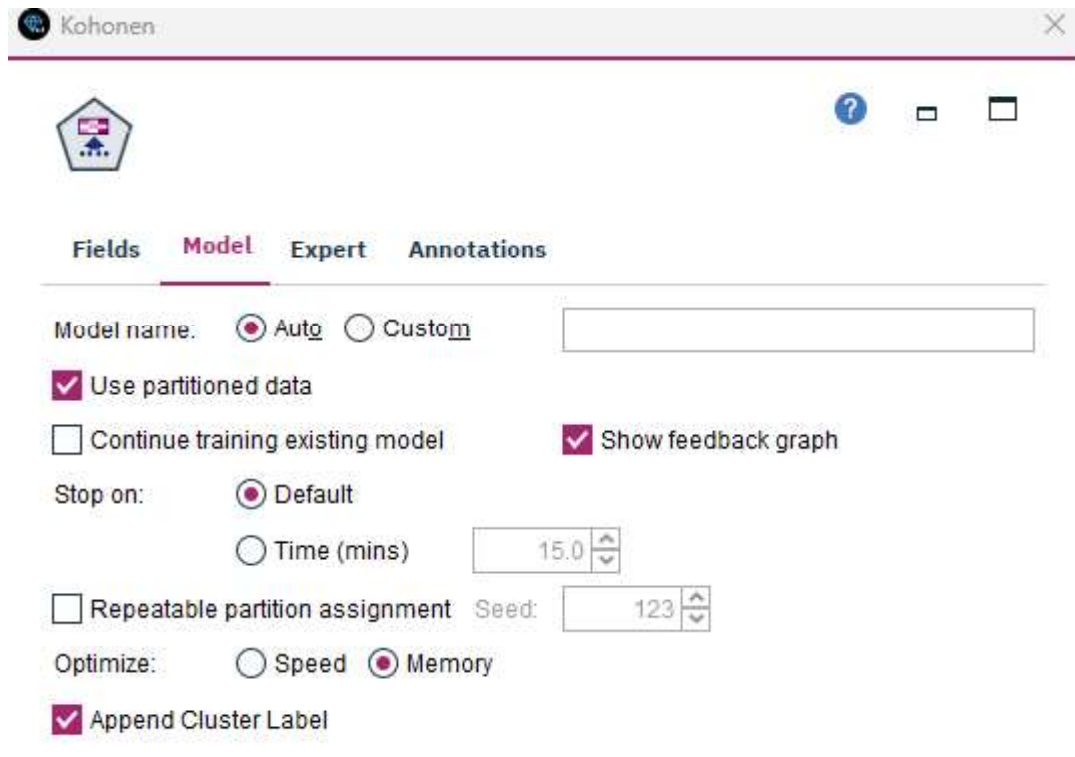
The feature importance chart showed that petal length and petal width were the most influential variables in determining cluster membership. These variables had the highest importance scores, indicating they play the largest role in separating the data into clusters.



The scatterplot for the TwoStep-AS model showed two distinct clusters. One cluster clearly separated a group of points, while the other contained overlapping values. This suggests that one species, likely Setosa, is clearly distinct, while the other two species are more similar.



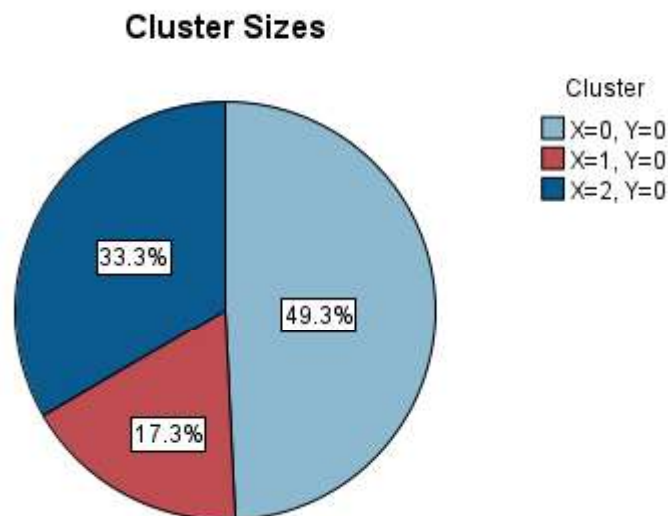
The Kohonen model was configured using standardized input variables.



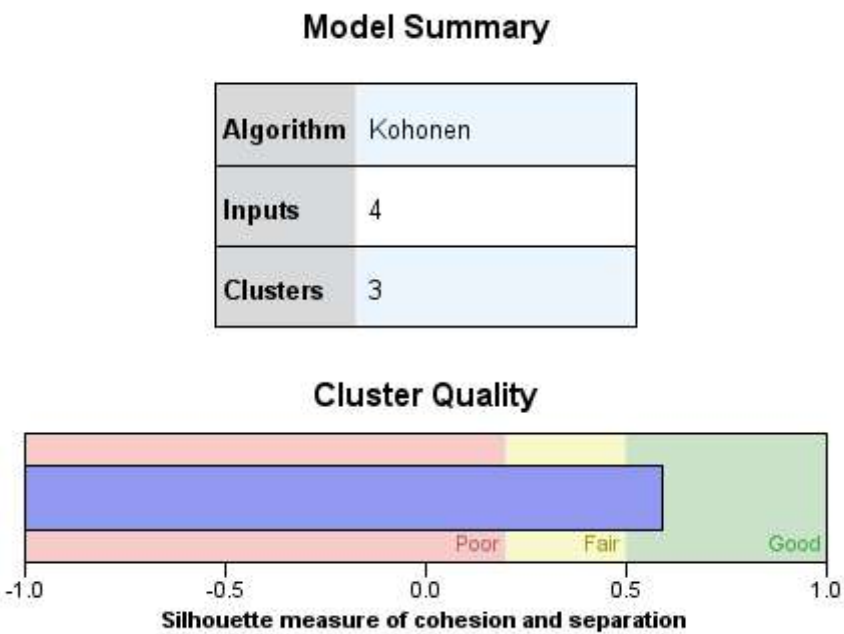
The screenshot shows the 'Kohonen' application window with the 'Model' tab selected. The configuration options are as follows:

- Model name: ☒ Auto ☐ Custom
- ☒ Use partitioned data
- ☐ Continue training existing model
- ☒ Show feedback graph
- Stop on: ☒ Default ☐ Time (mins) (15.0)
- ☐ Repeatable partition assignment Seed: 123
- Optimize: ☐ Speed ☒ Memory
- ☒ Append Cluster Label

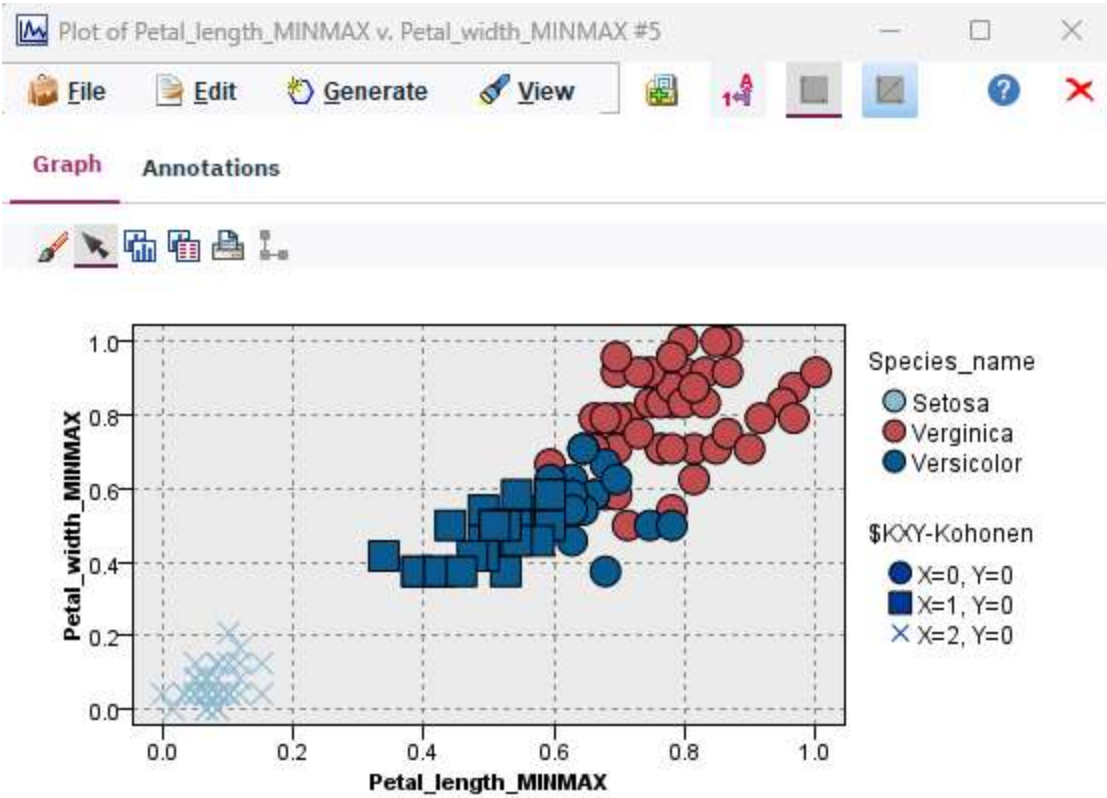
The model used partitioned data and was set to generate cluster labels. This neural network-based clustering method organizes data into clusters using self-organizing maps.



The Kohonen model produced three clusters. The cluster size chart showed three groups with varying proportions.

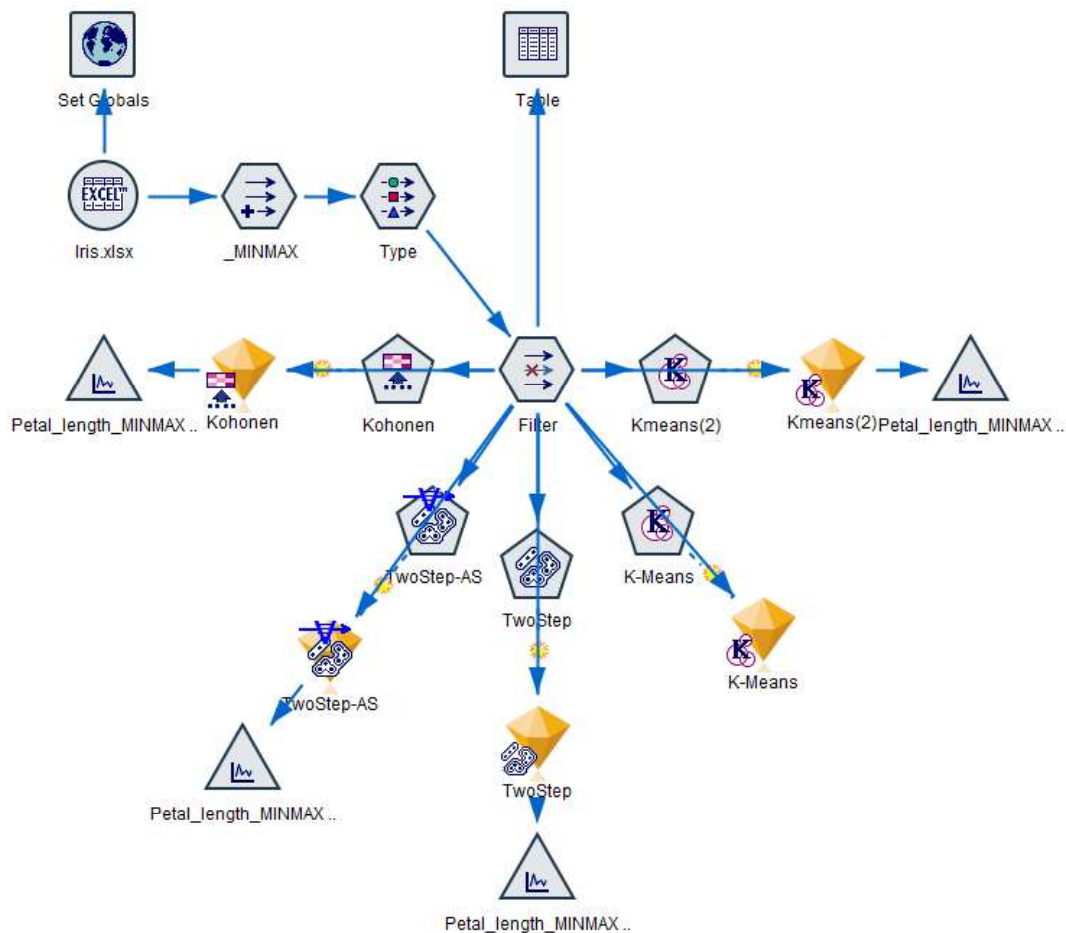


The model summary confirmed the use of four input variables and three clusters. The cluster quality measure indicated fair separation.



The Kohonen scatterplot showed three clusters visually. One cluster was clearly separated, while the other two showed some overlap. This again suggests that one species is distinct, while the remaining two are more similar in their measurements.

Results



Recommendations

Among the models evaluated, approaches that produced three clusters, such as K-Means with three clusters and the Kohonen model, are recommended when the goal is to reflect the natural structure of the dataset. Since the Iris dataset contains three known species, models that align with this structure provide more meaningful and interpretable results. In contrast, models

that produced only two clusters, such as the TwoStep model, tended to group Versicolor and Virginica together, which may be useful only in scenarios where a binary distinction is sufficient.

It is also recommended that data standardization continue to be used in future analyses. The MinMax normalization applied in this project ensured that all variables were on the same scale, which is critical for distance-based clustering methods like K-Means and Kohonen. Without this step, variables with larger numerical ranges could disproportionately influence cluster formation.

Although clustering provided useful insights into natural groupings within the data, there were limitations observed in separating Versicolor and Virginica. Future work should consider combining clustering with supervised learning techniques to improve classification accuracy. For example, after identifying clusters, a classification model could be trained to better distinguish between overlapping groups.

Finally, visual analysis using scatterplots proved highly effective in interpreting clustering results. It is recommended that visualization continue to be used alongside statistical measures such as cluster quality, as it provides a clearer understanding of how well the models are performing and where overlap occurs.

Supervised Learning

Objectives

Lastly, there is the Supervised Learning model. In this model, we used the BostonHousing.xlsx dataset. We used this dataset to configure this model in the IBM SPSS